Assignment 3 Feedback

25S2

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Q3(b) and (c)

X and Y are 2-bit values. Your answers to X and Y should be 2 bits.

Q4(c)

Note that for flip-flops, we always have both the flip-flop outputs and their complements readily available.

- $DC = A.C' + B'.C' = (A+B').C'$

- We need two logic gates to implement the OR and AND operators.

Q4(e) and (g)

The question asks for a simplest implementation of A^+ and B^+ using the minimum number of logic gates. No flip-flops are required here.

(e) $A^+ = E + (A \oplus B)$

- 2 gates

(g) $B^+ = (B+E)'$

- 1 NOR gate

Congratulations for completing your last assignment of CS2100!